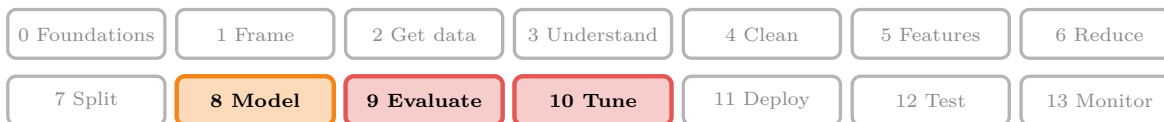


Bagging Regressor

Contents

1. Overview	2
2. The core idea	2
3. Seeing it on a curve	3
4. BaggingRegressor on the Boston housing data	4
4.1 Three single regressors	4
4.2 A bagging regressor with default settings	4
4.3 Tuning with GridSearchCV	5
4.4 The out-of-bag score	6
5. Summary	7
6. Sources	7
7. Key terms	7

Where this fits: Pipeline map steps 8 (Model), 9 (Evaluate), 10 (Tune). See the [Course map](#).



- **Builds on:** Simple imputation (mean, median, mode, constant) ([Note 37](#)); ML pipelines ([Note 38](#)); Bias-variance trade-off ([Note 62](#)); Overfitting ([Note 91](#)); Hyperparameter tuning ([Note 98](#)); Decision trees ([Note 100](#)).
- **Leads to:** Random forest ([Note 108](#)).
- **Compare with:** Boosting ([Note 101](#)); Voting ensembles ([Note 104](#)); Cross-validation ([Note 104](#)); Random forest ([Note 108](#)); Optuna ([Note 134](#)); Bayesian optimisation ([Note 134](#)).

1. Overview

Key point: A bagging regressor works exactly like a bagging classifier, except that aggregation takes the mean of the base models' numbers instead of a majority vote.

The bagging regressor applies bagging (the [bagging Note](#)) to regression. Everything carries over from the [bagging classifier Note](#): the four sampling types and every hyperparameter. This Note covers:

- the one difference, in the aggregation step;
- a demo with curves: one tree against a bagged ensemble;
- `BaggingRegressor` on the Boston housing data, tuned with `GridSearchCV`.

The Notebook (`notebook.ipynb`) runs every experiment. The Dash app `app.py` lets us change the base model and the bagging settings and redraws both curves.

2. The core idea

Key point: Bootstrapping is unchanged; aggregation averages the predictions.

In the **bootstrapping** step, each base model is trained on its own random sample of the data, as before. So pasting, random subspaces and random patches are all available too.

In the **aggregation** step, each base model returns a number, since this is regression. The bagging regressor returns their **mean**. Taking the mean is the only change.

scikit-learn's `BaggingRegressor` has exactly the same hyperparameters as `BaggingClassifier` (the [bagging classifier Note](#), section 6): `estimator`, `n_estimators`, `max_samples`, `max_features`, `bootstrap`, `bootstrap_features` and `oob_score`.

3. Seeing it on a curve

Key point: One fully grown tree passes through every training point (test R^2 0.92); 50 bagged trees give a smooth curve that follows the pattern (0.95).

The demo data has one **feature** (input variable, one column of the data table). Its **target** (the output we predict) is two Gaussian bumps (a small one at 0 and a larger one at 2) plus a little noise, with 150 training **observations** (points, one row of the table each) and 100 test points. With a single feature there are no features to sample, so the app only offers observation settings: the base model (decision tree, SVR or KNN), `n_estimators`, `max_samples` and `bootstrap`.

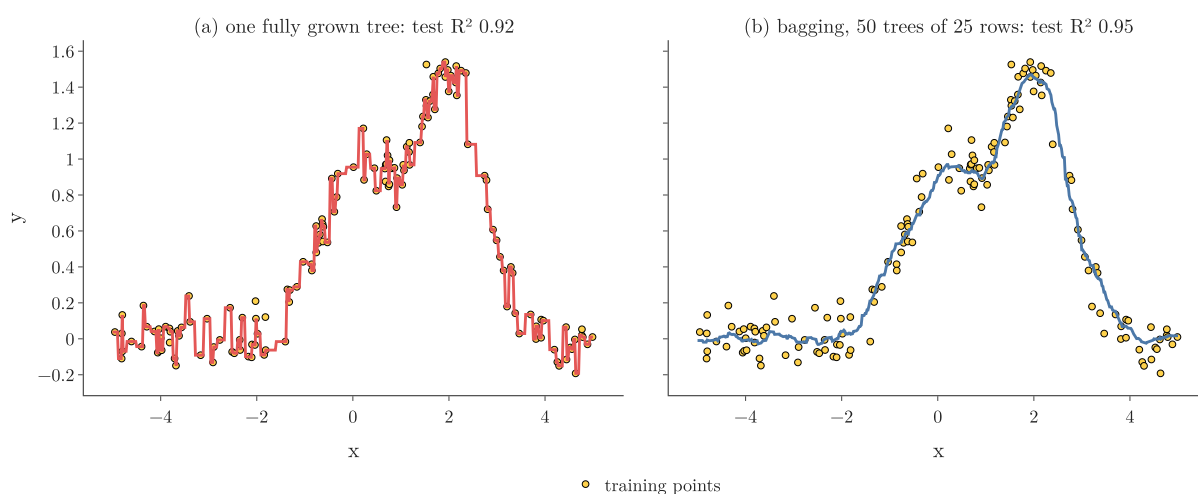


Figure 1: (a) One fully grown regression tree; (b) a bagging regressor of 50 trees, each trained on 25 observations drawn with replacement

Figure 1 compares a single decision tree with bagging (50 trees, 25 observations each, with replacement):

- **(a) The single tree** tries to pass through every training point, outliers included: low bias, high variance, so it overfits. Test R^2 : **0.92**.
- **(b) The bagging regressor** gives a smooth curve. The curve does not touch every point; it follows the underlying pattern. Test R^2 : **0.95**.

The smoothing is the same effect as in classification: bagging keeps the low bias and removes much of the variance.

Changing the settings makes little difference on this easy data: 100 trees score 0.952, and pasting (75 observations, `bootstrap=False`) scores 0.947.

Extra: Bagging helps unstable models, whose fit swings with small changes in the data, such as trees; it can slightly hurt stable ones, such as nearest-neighbour methods

(Breiman, 1996, sections 1 and 6.3; the [bagging classifier Note](#), section 2.2). The bumps data agrees: bagged KNN (0.832) and bagged SVR (0.871) both do worse than a single KNN (0.954) or SVR (0.894).

4. BaggingRegressor on the Boston housing data

Key point: Out of the box, bagged trees (mean test R^2 0.84) beat linear regression (0.71), a single tree (0.72) and KNN (0.50).

4.1 Three single regressors

Key point: Averaged over 100 random splits: linear regression 0.71, decision tree 0.72, KNN 0.50.

The Boston housing data (the [regression trees Note](#), section 7.1) has 506 districts (observations), 13 features and the median house price as the target. We split it into 404 training observations and 102 test observations, and train three single regressors.

A test set of 102 observations is small, so one split gives a noisy score: on our first split, for example, the single tree scores only 0.38. So the Notebook repeats the split 100 times, with a different random shuffle each time, and every number in this section is the mean over those 100 splits:

Model	Linear regression	Decision tree	KNN
Mean test R^2	0.708	0.716	0.504

4.2 A bagging regressor with default settings

Key point: Ten bagged trees, nothing tuned: mean test R^2 0.84, above every single model.

Python: A bagging regressor with every setting at its default.

```
from sklearn.ensemble import BaggingRegressor
```

```
bag = BaggingRegressor(random_state=1)
bag.fit(X_train, y_train)
bag.score(X_train, y_train)    # 0.980 (first split)
bag.score(X_test, y_test)     # 0.828 (first split)
```

The defaults: `estimator=None` (a decision tree), `n_estimators=10`, as many observations per model as the training set (`max_samples=None`), drawn with replacement, and every feature. `score` returns R^2 for a regressor.

Without tuning anything, the bagging regressor scores a mean test R^2 of **0.841** over the 100 splits, well above all three single models (0.708, 0.716 and 0.504). The ten trees are each

as unstable as the single tree, but their average is far steadier. On the first split, the gap between training (0.98) and test (0.83) shows some overfitting remains.

4.3 Tuning with GridSearchCV

Key point: A grid search over 144 combinations picks 50 bagged trees on full bootstrap samples; over the 100 splits these settings score 0.857, above the default's 0.841.

Instead of trying pasting, random subspaces and random patches by hand, we let GridSearchCV try them all (the [KNN Note](#), section 4.2):

Python: Searching over the base model and the sampling settings.

```
params = {
    "estimator": [None, LinearRegression(),
                  KNeighborsRegressor()],
    "n_estimators": [20, 50, 100],
    "max_samples": [0.5, 1.0],
    "max_features": [0.5, 1.0],
    "bootstrap": [True, False],
    "bootstrap_features": [True, False]}
grid = GridSearchCV(BaggingRegressor(random_state=1),
                    param_grid=params, cv=3,
                    n_jobs=-1, verbose=1)
grid.fit(X_train, y_train)
grid.best_params_
```

The base model itself is a hyperparameter here: `None` stands for the default decision tree. With `bootstrap` and `bootstrap_features` both tried as `True` and `False`, the search covers bagging, pasting, random subspaces and random patches.

The grid holds $3 \times 3 \times 2 \times 2 \times 2 \times 2 = 144$ combinations, each with 3-fold cross-validation: 432 fits, under a minute with `n_jobs=-1`. The best settings:

- base model: **decision tree**;
- `bootstrap=True`: **bagging**, not pasting;
- `bootstrap_features=False`, `max_features=1.0`: **no feature sampling**;
- `max_samples=1.0`: every model gets as many observations as the training set;
- `n_estimators=50`.

Their cross-validation R^2 is **0.871**. On the same 100 splits as above, these settings score a mean test R^2 of **0.857**, close to the cross-validation estimate and above the default bagging regressor (0.841); they beat the default on 79 of the 100 splits.

Model	Linear regression	Decision tree	KNN	Bagging, default	Bagging, tuned
Mean test R^2 (100 splits)	0.708	0.716	0.504	0.841	0.857

Extra: One split can mislead. On the first split alone, the tuned model scores 0.806 and the default 0.828, the opposite order. Over the 100 splits, the tuned settings score anywhere from 0.63 to 0.94, so one 102-observation test score can land far from the average. The search picks settings that do best on average, which is what the mean over many splits measures. Expecting one test score to match `best_score_` exactly is a common mistake.

4.4 The out-of-bag score

Key point: With `oob_score=True`, the regressor reports R^2 on the out-of-bag observations: 0.870 here.

`oob_score` works for regression too (the [bagging classifier Note](#), section 4). For a regressor, `oob_score_` is the R^2 on the observations each model never saw. With 50 trees on the first split, the OOB R^2 is **0.870**, close to the cross-validation score.

5. Summary

	Bagging classifier	Bagging regressor
Base models	classifiers	regressors
Aggregation	majority vote	mean
scikit-learn	<code>BaggingClassifier</code>	<code>BaggingRegressor</code>
score and <code>oob_score_</code>	accuracy	R^2
Hyperparameters	<code>estimator</code> , <code>n_estimators</code> , <code>max_samples</code> , <code>max_features</code> , <code>bootstrap</code> , <code>bootstrap_features</code> , <code>oob_score</code>	the same

- Only aggregation changes for regression: the mean replaces the vote.
- On the bumps data, bagging smooths one overfitting tree into a curve that follows the pattern: test R^2 0.92 to 0.95.
- On Boston (mean over 100 splits), default bagged trees score 0.84 against 0.72 for the best single model; a grid search picks plain bagging of 50 trees, which scores 0.86.
- Bagging helps unstable models such as trees; bagged KNN and SVR do worse than single ones.

6. Sources

- Breiman, L. (1996). “Bagging Predictors”. *Machine Learning* 24(2), 123–140, sections 1 and 6.3.

7. Key terms

Term	Meaning
Observation	One record of the data: one row of the data table
Feature	An input variable: one column of the data table
Target	The output we predict
Bagging regressor	A bagging ensemble of regressors that predicts the mean of their predictions
<code>BaggingRegressor</code>	scikit-learn class for bagging, pasting, random subspaces and random patches in regression